Homework 7 (Simplex Method Variations)

March 5, 2026

For this homework, each question is worth 10 points.

1)

Let $f(n) = O(n^p)$ and $g(n) = O(n^q)$. Suppose $c, d \neq 0$ are real numbers. Then show

$$cf(n) + dg(n) = O\left(n^{\max\{p,q\}}\right)$$

2)

a)

Let C be an invertible $m \times m$ matrix and let $\vec{u}, \vec{v} \in \mathbb{R}^m$. Show

$$\left(C + \vec{u}\vec{v}^T\right)^{-1} = C^{-1} - \frac{C^{-1}\vec{u}\vec{v}^T C^{-1}}{1 + \vec{v}^T C^{-1}\vec{u}}$$

(Hint: $\vec{v}^T C^{-1}\vec{u}$ is a scalar and matrix multiplication is associative).

b)

Assuming you are given C^{-1}, show that $\left(C + \vec{u}\vec{v}^T\right)^{-1}$ can be obtained in $O\left(m^2\right)$ multiplication and division operations. (Hint: Associativity may help here. Naive matrix multiplication is $O\left(n^3\right)$, but some special cases, like matrix/vector multiplication or the outer product of two vectors are $O\left(n^2\right)$).

3)

Suppose you are using the Tableau implementation of the simplex method. Midway through the process, you end up with the following tableau:

$$\begin{bmatrix} & * & x_1 & x_2 & x_3 & x_4 & x_5 \\ & -10 & \delta & 2 & 0 & 0 & 0 \\ x_3 = & 4 & -1 & \eta & 1 & 0 & 0 \\ x_4 = & 1 & \alpha & -4 & 0 & 1 & 0 \\ x_5 = & \beta & \gamma & 3 & 0 & 0 & 1 \end{bmatrix}$$

(The 1st row starts with x_3 and the 1st column starts with $\vec{x}_1$). For each of the cases below, find values of $\alpha, \beta, \gamma, \delta, \eta$ to make the statement true.

a)

The current solution is optimal and there are multiple distinct optimal solutions to the LP.

b)

The optimal cost is $-\infty$.

c)

The current solution is feasible but not optimal.

4)

Let $\vec{x}, \vec{y} \in \mathbb{R}^n$ and let $\vec{x} \overset{L}{<} \vec{y}$ if $\vec{x}$ is before $\vec{y}$ in the lexicographic ordering.

a)

Sketch the following set:

$$S = \left\{ \vec{x} \in \mathbb{R}^2 | \begin{pmatrix} 1 \\ 0 \end{pmatrix} \overset{L}{<} \vec{x} \overset{L}{\leq} \begin{pmatrix} 2 \\ 1 \end{pmatrix} \right\}$$

b)

Determine if S is convex. If S is convex, prove it. Otherwise, show why it isn't.

5)

Prove the following:

a)

Suppose $\vec{x} \overset{L}{<} \vec{y}$. Prove that $\vec{x} + \vec{z} \overset{L}{<} \vec{y} + \vec{z}$ for all $\vec{z} \in \mathbb{R}^n$

b)

Suppose $\vec{x} \overset{L}{<} \vec{y}$ and suppose $\lambda \in \mathbb{R}$ such that $\lambda > 0$. Prove that $\lambda\vec{x} \overset{L}{<} \lambda\vec{y}$.

6)

How is this class going so far? Do you have any questions, comments, concerns, or feedback at this point?